

Homological Algebra Lecture Notes

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1 Lecture 1 (October 1st 2026)

Preview: We basically spend some time doing category theory, and the theory of functors with some attention paid to R -Modules. There is some additional mastery material that we need to cover as well.

Basics of Category Theory

Definition 1.1. A *category* $\mathcal{C}$ consists of

1. a collection of ‘objects’ called $\text{ob}(\mathcal{C})$
2. $\forall A, B \in \text{ob}(\mathcal{C})$ we have a set of “morphisms” $\text{Hom}_{\mathcal{C}}(A, B)$
3. An identity (or a unit) for each object $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$
4. A composition rule for A, B, C which is: $\text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$ It is simple to think of $g \circ f \in \text{Hom}_{\mathcal{C}}(A, C)$ where $g \in \text{Hom}_{\mathcal{C}}(B, C)$ and $f \in \text{Hom}_{\mathcal{C}}(A, B)$ so that we really have composition of functions here.
5. These functions are also associative such that for $f \in \text{Hom}_{\mathcal{C}}(A, B), g \in \text{Hom}_{\mathcal{C}}(B, C), h \in \text{Hom}_{\mathcal{C}}(C, D)$ we have $h \circ (g \circ f) = (h \circ g) \circ f$ and the same for units $\text{id}_B \circ f = f \circ \text{id}_A = f$.

| * Aside: add intuition here

Example 1.2. The category of sets $\mathcal{C}$ can be considered and the morphisms on it are $\text{Hom}_{\mathcal{C}}(A, B)$ where A, B are sets. This naturally gives us $\text{Hom}_{\mathcal{C}}(A, B) = \text{Func}(A, B)$.

Example 1.3. The category of groups. $\mathcal{C} = \text{groups}$ and we form the category of groups. We initially have $\text{ob}(\text{Groups})$ and $\text{Hom}_{\text{Groups}}(A, B) =$ the set of group homomorphisms between groups A and B .

Example 1.4. The category of topological spaces. $\mathcal{C} = \text{Top}$ the category of topological spaces so we have. $\text{ob}(\text{Top}) =$ all topological spaces, and then we also have $\text{Hom}_{\text{Top}}(A, B) =$ all continuous functions between A and B .

For the next example we introduce rings and modules. R is a ring (not necessarily commutative) and it has modules say A, B satisfying the usual axioms.

Example 1.5. The category of R -Modules. $\mathcal{C} = R\text{-Modules}$ where, for example, $A, B \in \text{ob}(R\text{-Modules})$ and we have $\text{Hom}_R(A, B)$ as the R -Module homomorphisms between modules A and B .

Now, we move onto subcategories which is exactly what you think it is.

Definition 1.6. A *subcategory* of a category, $\mathcal{D} \subseteq \mathcal{C}$ is a subcollection $\text{ob}(\mathcal{D}) \subseteq \text{ob}(\mathcal{C})$ and we have for all $X, Y \in \text{ob}(\mathcal{D})$ we have $\text{Hom}_{\mathcal{D}}(X, Y) \subseteq \text{Hom}_{\mathcal{C}}(X, Y)$ such that

1. $\text{id}_X \in \text{Hom}_{\mathcal{D}}(X, X)$
2. It is closed under composition (Need more substance here).

We call $\mathcal{D}$ a full subcategory of $\mathcal{C}$ if $\forall X, Y \in \text{ob}(\mathcal{D})$ we have $\text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y)$.

Example 1.7. The category of abelian groups. We will call this $\mathcal{D} = \text{Ab}$. We have $\text{Ab} \hookrightarrow \text{Groups}$ is a full subcategory since the fact that two groups are abelian doesn't change the homomorphisms between them. Hence, $\text{Hom}_{\text{Ab}}(A, B) = \text{Hom}_{\text{Groups}}(A, B)$ for any choice $A, B \in \text{ob}(\text{Ab})$.

Example 1.8. For a field k , we consider the category of vector spaces of k , $\text{Vect}_k = \mathcal{C}$, then we can have the sub-category of finite dimensional vector spaces of k , $\text{fd}(\text{Vect}_k) = \mathcal{D}$, $\mathcal{D} \subseteq \mathcal{C}$ is a full subcategory.

Now, moving ahead with a view towards functors, we need to introduce some machinery to understand some properties we will learn about later like contravariance.

Definition 1.9. if $\mathcal{C}$ is a category then $\mathcal{C}^{\text{op}}$ is the opposite category defined such that $\text{ob}(\mathcal{C}) = \text{ob}(\mathcal{C}^{\text{op}})$ where $\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathcal{C}}(Y, X)$.

Definition 1.10. A (covariant) *functor* on two categories $\mathcal{C}$ and $\mathcal{D}$, $F : \mathcal{C} \rightarrow \mathcal{D}$ is an assignment between $\text{ob}(\mathcal{C}) \rightarrow \text{ob}(\mathcal{D})$ that takes $X \in \text{ob}(\mathcal{C})$ to $F(X) \in \text{ob}(\mathcal{D})$ so we have F taking $X \rightarrow F(X)$ such that for any $X, Y \in \text{ob}(\mathcal{C})$ we know there exists $F_{X,Y} : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$ where $F(f) : F(X) \rightarrow F(Y)$ which also must have the properties that $F(\text{id}_X) = \text{id}_{F(X)}$ and that $F(f \circ g) = F(f) \circ F(g)$.

* **Aside:** Ok, this is really abstract and frankly warrants a commutative diagram to describe what functors can actually do. We start with $F : \mathcal{C} \rightarrow \mathcal{D}$ which has two assignments inside of it.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ F \downarrow & & \downarrow F \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

Thus, we can see that F can take in the morphism between $F(X)$ and $F(Y)$ which means we similarly have

$$F : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$$

The final part of our definition is just making sure it plays nicely with those morphisms.

Definition 1.11. A *contravariant functor* is a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. This means it flips the direction when we apply it to spaces and functions between their homomorphisms $\text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(Y), F(X))$.

* **Aside:** Here, I will try to make another diagram to illustrate how a contravariant functor works. We start with a contravariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ which has two assignments,

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ F \downarrow & & \downarrow F \\ F(X) & \xleftarrow{F(f)} & F(Y) \end{array}$$

This makes it evident to see that we have the opposite direction only on the bottom

$$F : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(Y), F(X))$$

This is because the morphisms over $\mathcal{C}^{\text{op}}$ are $\text{Hom}_{\mathcal{C}}(Y, X)$.

Example 1.12. Here we will let $\mathcal{C} = \text{Top}$ and $\mathcal{D} = \text{Set}$ and we take the natural function from a topological space to the set underneath it. $F : \text{Top} \rightarrow \text{Set}$ which forgets the topology and just views the former topological space as a set. The precise way to think of this is that

1. $F(X)$ is merely the set X where earlier it had a topology.
2. $f \in \text{Hom}_{\text{Top}}(X, Y)$ is continuous since all the morphisms are continuous but $F(f)$ is merely the function, it forgets that it is continuous.

Definition 1.13. A functor $F, F : \mathcal{C} \rightarrow \mathcal{D}$ is called fully faithful if $F : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$ is bijective for X, Y in $\mathcal{C}$

From now on $\text{ob}(\mathcal{C})$ will be replaced by just $\mathcal{C}$.

Example 1.14. For any category $\mathcal{C}$ and any object $A \in \mathcal{C}$ there is a contravariant functor $\mathcal{C} \rightarrow \text{Set}$ where $F = (\text{Hom}_{\mathcal{C}}(-, A))$ where the dash is where we will take an input for F . Hence, for $B \in \mathcal{C}$ we have $F(B) = \text{Hom}_{\mathcal{C}}(B, A)$ and for $f \in \text{Hom}_{\mathcal{C}}(B, B')$ we get $F(f) \in \text{Hom}_{\mathcal{C}}(F(B'), F(B))$. This is a function $\text{Hom}_{\mathcal{C}}(B', A) \rightarrow \text{Hom}_{\mathcal{C}}(B, A)$. Functors like this example are called representable and are represented by A .

*** Aside:** Here we need to add a commutative diagram for 1.14 to make sure it is obvious what is going on. Placeholder.

Definition 1.15. Let $\mathcal{C}$ and $\mathcal{D}$ be categories with functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$. A *natural transform* $\eta : F \rightarrow G$ is for each $C \in \mathcal{C}$ a morphism $\eta_C : F(C) \rightarrow G(C)$ such that for any morphism $f : C \rightarrow C'$ both $C, C' \in \mathcal{C}$ the following diagram commutes.

$$\begin{array}{ccc} F(C) & \xrightarrow{\eta_C} & G(C) \\ F(f) \downarrow & & \downarrow G(f) \\ F(C') & \xrightarrow{\eta_{C'}} & G(C') \end{array}$$

An easy example of our first natural transformation can come from our earlier functors on topological spaces and sets. In particular there are two ways you can pretty easily put a topology on a set: make it the discrete topology, and make it the trivial topology.